Applying Artificial Intelligence to User Experience Research : Leveraging Machine Learning to Interpret Quantitative User Experience Evaluation Questionnaire Results

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ABSTRACT

User experience is today a key dimension in the design and development of services, as it enables the creation of lasting connections with users. This field has led to the development of a wide variety of interactive interfaces, ranging from websites and mobile applications to complex interactive systems. Several techniques exist to evaluate the effectiveness of these interfaces, and they often result in quantitative data that still require detailed interpretation in order to understand the specific changes that need to be made to the system. This literature review aims to examine existing methods designed to facilitate the interpretation of quantitative UX evaluation results for researchers, with a particular focus on approaches leveraging artificial intelligence technologies, more specifically machine learning.

Author Keywords

User Experience; UX Evaluation; Quantitative Questionnaires; Machine Learning; Artificial Intelligence; Interface Design

INTRODUCTION

User experience refers to the quality of the interaction between a user and a system through an interface. It is a criterion that must be carefully considered during product development, as it has a significant impact on the success of the product, notably by helping to attract and retain users. By interface, we refer to any artifact that mediates the interaction between a human and a machine. A wide range of interfaces can illustrate this definition, such as a website page, the digital control panel of a printer, or the controller of a video game console. The development of such interfaces follows several stages. First, an "empathy" phase aims to place the developer in the user's position in order to understand their needs when interacting with the system. These needs are then defined and implemented in a prototype, which is subsequently tested, a crucial step in the interface development process. Indeed, this phase makes it possible to identify in detail which elements need to be modified in order to improve the overall experience. Testing focuses on the evaluation of several aspects of the system, particularly ergonomic and aesthetic qualities. Evaluation techniques are diverse. They include conducting interviews to directly gather users' opinions about the system, asking users to perform tasks on the interface to assess its usability, often relying on metrics such as task completion time, and the use of questionnaires. Each of these techniques enables a more or less detailed evaluation depending on the type of interface being tested. In the context of this study, we focus on the use of questionnaires. While questionnaires are an effective method for evaluating interfaces, their use also requires the ability to translate often numerical results, such as averages, into concrete interface elements that need to be modified. For this reason, this literature review explores whether and to what extent machine learning and artificial intelligence technologies are currently used to support the interpretation of data produced by UX evaluation questionnaires. More specifically, it examines existing research that investigates the use of such techniques to relate quantitative questionnaire results to specific interface design elements, such as typography, colors, or layout. The review is structured around three key aspects of the topic : the different types of UX evaluation questionnaires, existing approaches to the interpretation of questionnaire results, and the application of machine learning techniques within the field of user experience. To guide this literature review, we ask : To what extent are machine learning and artificial intelligence techniques currently applied to interpret quantitative UX evaluation questionnaire results, and what approaches have been explored in the literature ?

METHOD

The initial search focused on identifying papers related to the field of user experience (UX) evaluation. The search was primarily conducted using the IEEE Xplore Digital Library, which provides advanced filtering and search capabilities for precise queries. Based on the general domains linked to our study, we defined these specific fields and conducted separate searches for each one of them.

Our first search targeted papers addressing user evaluation techniques, using the query : `("Abstract": "user experience questionnaire") OR ("Abstract": "usability scale")`. This returned a total of 3,667 papers. We then filtered these results to include only papers

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related to machine learning using the built-in IEEE Xplore filters, reducing the set to 125 papers. An additional search focusing on the interpretation of UX questionnaire results was conducted with : `("Abstract": "user experience") AND ("Abstract": "interpretation"`, yielding 23 papers.

To ensure the quality and relevance of the dataset, we applied the following inclusion and exclusion criteria :

- Only journals or conference papers were included, while books, magazine articles, standards, and courses were excluded. We also applied a time filter, retrieving papers oublished between 2019 and 2026. This reduced the first search to 2,580 papers, the machine learning subset to 112, and the interpretation subset to 19 papers. Note that we included a few older, well-established papers on classic questionnaire techniques, such as the System Usability Scale, to ensure that the review provides a comprehensive overview of the field.

- Papers had to directly address the development or assessment of UX evaluation techniques. Articles that only used questionnaires to evaluate specific systems were excluded, as our focus is on the methodology and scientific principles behind the questionnaires themselves, along with the methods used to interpret the results.

- For the machine learning subset, we retained only papers that applied machine learning or artificial intelligence techniques specifically to assess user experience of a system. Articles using these technologies for unrelated tasks were excluded.

After applying these criteria, the final dataset consisted of 11 primary studies, which were then analyzed in detail in this literature review. These papers include works on UX questionnaire design (e.g., SUS, UEQ, AttrakDiff), studies on interpreting questionnaire results, and research applying machine learning techniques to UX evaluation.

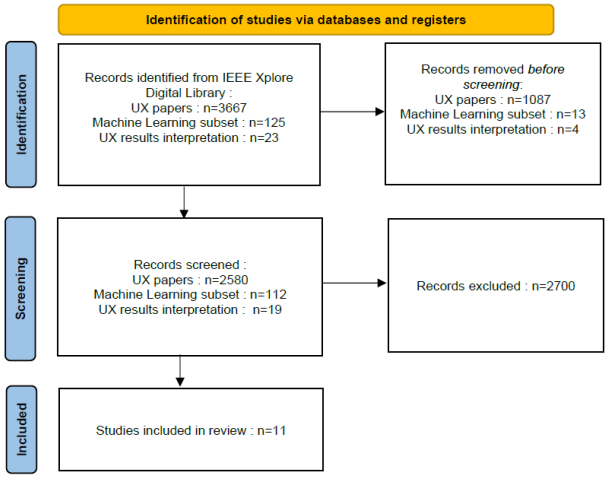


Figure 1. PRISMA Paper Research Diagram

RESULTS

As mentioned in the previous section, our literature research is focused on three main aspects :

- Questionnaire-based UX evaluation methods
- Methods for interpreting questionnaires results
- Use of machine learning in UX evaluation

Several studies describe and rely on standardized questionnaires that focus on users subjective perceptions of an interface rather than its technical implementation. Among the most frequently cited instruments is the System Usability Scale (SUS)[2], which provides a global usability score based on users ratings of predefined statements. The SUS is widely adopted due to its simplicity and its ability to generate quantitative measures such as overall scores, means, and standard deviations. Another commonly used instrument is the User Experience Questionnaire (UEQ)[10], which evaluates multiple dimensions of user experience, including attractiveness, efficiency, stimulation, and novelty. The UEQ allows researchers and practitioners to obtain a multidimensional assessment of an interface through aggregated numerical scores. The AttrakDiff[11] questionnaire is also prominently discussed in the literature. It focuses on pragmatic and hedonic qualities of interaction by asking users to rate pairs of opposite adjectives. The results are typically summarized using category means and visualized through diagrams or portfolios, which help position an interface along predefined quality dimensions. Overall, the literature highlights that questionnaire-based evaluations primarily produce quantitative outcomes, which serve as the main input for subsequent analysis and decision-making.
Beyond these widely used standardized instruments, the literature also highlights the emergence of domain-specific questionnaires tailored to particular contexts of use. For example, recent work has proposed a dedicated questionnaire for evaluating user experience in virtual reality applications within healthcare settings[3]. Such instruments are designed to capture dimensions that are not adequately addressed by generic questionnaires, including presence, realism, emotional state, or side effects specific to immersive environments. Despite their contextual specialization, these questionnaires similarly rely on Likert-scale items and produce quantitative scores that must be interpreted to inform design decisions.

While questionnaires generate numerical scores, translating these values into concrete interface improvements remains a complex task. Most studies indicate that interpretation is traditionally performed by human experts, who combine statistical results with contextual information and qualitative user feedback. This process often relies on domain knowledge and subjective judgment.[7] Only a limited number of research works explicitly address systematic methods for interpreting questionnaire data. One identified approach proposes mapping UEQ results onto a two-dimensional graph combining performance and importance, allowing characteristics to be categorized into different priority zones (e.g., areas requiring improvement or potential overdevelopment)[6]. Although such methods help structure interpretation at a higher level, the literature indicates that

they often lack precision when it comes to identifying specific interface elements that should be modified. As a result, interpretation remains a bottleneck in the UX evaluation process.

Withthe emergence of machine learning as a research field, its integration into other domains has been widely encouraged, including within the field of user experience. Prior work highlights the potential of machine learning techniques to support UX-related tasks. However, existing applications focus on the design and adaptation phases of user interfaces rather than on the evaluation phase. [1]. Several studies explore how machine learning can be used to analyze user behavior, recognize emotional states or adapt interfaces based on user preferences. Commonly reported approaches include the use of classification and prediction models such as random forests, decision trees, neural networks, and bayesian models. These techniques are shown to be effective in recognizing user emotions, predicting satisfaction, or supporting adaptive interface design.[4] [8] However, the literature reveals that few studies directly focus on applying machine learning to questionnaire-based UX evaluation data. Existing work in this area primarily investigates the prediction of overall user satisfaction from questionnaire responses, demonstrating high predictive accuracy in some cases.[9] One notable exception is a study that applies artificial neural networks to data collected using the User Experience Questionnaire (UEQ). In this work, neural network models are trained on questionnaire responses to estimate the relationships between different UX dimensions without relying on traditional statistical assumptions such as linearity or normality. The results reveal asymmetric and nonlinear relationships between application attributes and UX dimensions, particularly those related to usability. While this approach demonstrates the relevance of machine learning for analyzing questionnaire-based UX data, it remains focused on high-level UX dimensions and does not directly support the identification of specific interface elements requiring redesign. [5]

While these studies suggest that machine learning can successfully process questionnaire data, they remain limited in scope and do not yet fully address the interpretation of questionnaire results at a granular interface level.

DISCUSSION

The literature reviewed in this study shows that standardized quantitative questionnaires remain the most widely used tools for evaluating user experience. Instruments such as SUS, UEQ, and AttrakDiff provide valuable and reliable indicators to assess the overall quality of an interface and its key experiential dimensions. However, while these questionnaires offer important high-level insights, they often require additional effort to translate numerical results into concrete and actionable design decisions. In practice, understanding what exactly does not work within a specific interface still largely depends on expert interpretation and complementary qualitative methods. Recent research highlights a growing interest in applying machine learning techniques within the field of user experience, particularly to analyze user behavior, emotional responses, or overall satisfaction. Although these approaches contribute to a better understanding of users perceptions and opinions, they remain mostly focused on indirect signals or aggregated outcomes. To date, there is limited work addressing the use of machine learning specifically for the interpretation of quantitative UX questionnaire results in a way that directly links scores to concrete interface elements. This reveals a significant research gap at the intersection of machine learning and UX evaluation, where automated or semi-automated interpretation of questionnaires data could meaningfully support designers and researchers without relying exclusively on direct user feedback.

CONCLUSION

This literature review examined existing research on the use of quantitative UX evaluation questionnaires and the methods employed to interpret their results, with particular attention to emerging applications of machine learning in the field of user experience. The studies confirm that standardized questionnaires are widely adopted and provide robust quantitative indicators of interface quality. However, the literature also highlights that interpreting these numerical outcomes into precise and actionable design recommendations remains a challenging and largely manual process. While machine learning techniques are increasingly explored to analyze user behavior and emotions, their application to the direct interpretation of questionnaire-based UX evaluation data is still limited. Future work could focus on the development of methods and models based on machine learning techniques that take as input both the quantitative results of UX evaluation questionnaires and the specific interface elements to be modified. Such approaches could provide valuable support for UX designers and researchers, while also contributing to a deeper understanding of how to design more effective and user-centered interfaces.

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Explain how generative AI has been used in the assignment:

For each instance of use (see examples above), write an entry below and explain
1) *Purpose of use (what did you use the tool for?)*
2) *Which tool did you use? Include the link if possible, e.g., [Copilot with enterprise data protection (UCPH license), https://copilot.microsoft.com]*
3) *Work phase or step (when in the process did you use GAI?)*
4) *What did you do with the output? (including any editing of or continued work on the output)*
5) *What did you benefit of the tool? (think about the usefulness of the tool and reflect on your experience of using it, the applicability of the tool for this use, and your skill development in using GAI tools)*

1) I used ChatGPT to reformulate some technical words and concepts from the articles I read, that were not very clear to me. I also provided ChatGPT with some phrases and parts of my work where I wasn't sure of the translation and the overall structure and grammar.
2) ChatGPT
3) Mainly at the end when I re-read my work and wanted to make sure that the text was grammatically correct and clear.
4) I usually copy the output, read it and modify it to make it more fluid with the rest of my text.
5) The tool allow me to have a better understanding of some concepts and also to improve my English writing.

Document how generative AI has been used in the assignment:

To the extent possible, you must document your interaction with a GAI tool and the generated results, for example, in the form of appendices with prompts and results, and by citing (source referencing) GAI in your assignment. This means that it must be clear from your assignment which GAI tools you have used and how you have used the output of the GAI.

Please note: Content generated by GAI that is used as a source in the paper requires correct use of quotation marks and source referencing. Read the guidelines from Copenhagen University Library at KUnet here.

Figure 3. Declaration AI Page 2

Declaration, explanation, and documentation of using generative AI tools in ATHCC

Declaration of using generative AI tools (for students of ATHCC)

Declare here, that

☒ **I have used generative AI as an aid/tool** *(please tick)*

or

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However, the purpose of this GAI declaration is *not* to detect cheating. The intention is to integrate the use of these new technologies into current academic practices and support methodological reflections on the technology's contributions. Therefore, it is rather encouraged that you will learn to benefit from these tools in your assignment work and all future work!

Examples of GAI use:

- *Initial idea generation*
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- *Summarising large amounts of information*
- *Help with structure and outline*
- *Source and literature search*
- *Data processing (e.g. transcription, cleaning and structuring data)*
- *Data analysis*
- *Grammar check*
- *Feedback / review tool for revising text*
- *Grammar / language check tool*
- *As object of the analysis*

Figure 2. Declaration AI Page 1